

Lesson 14: Variance

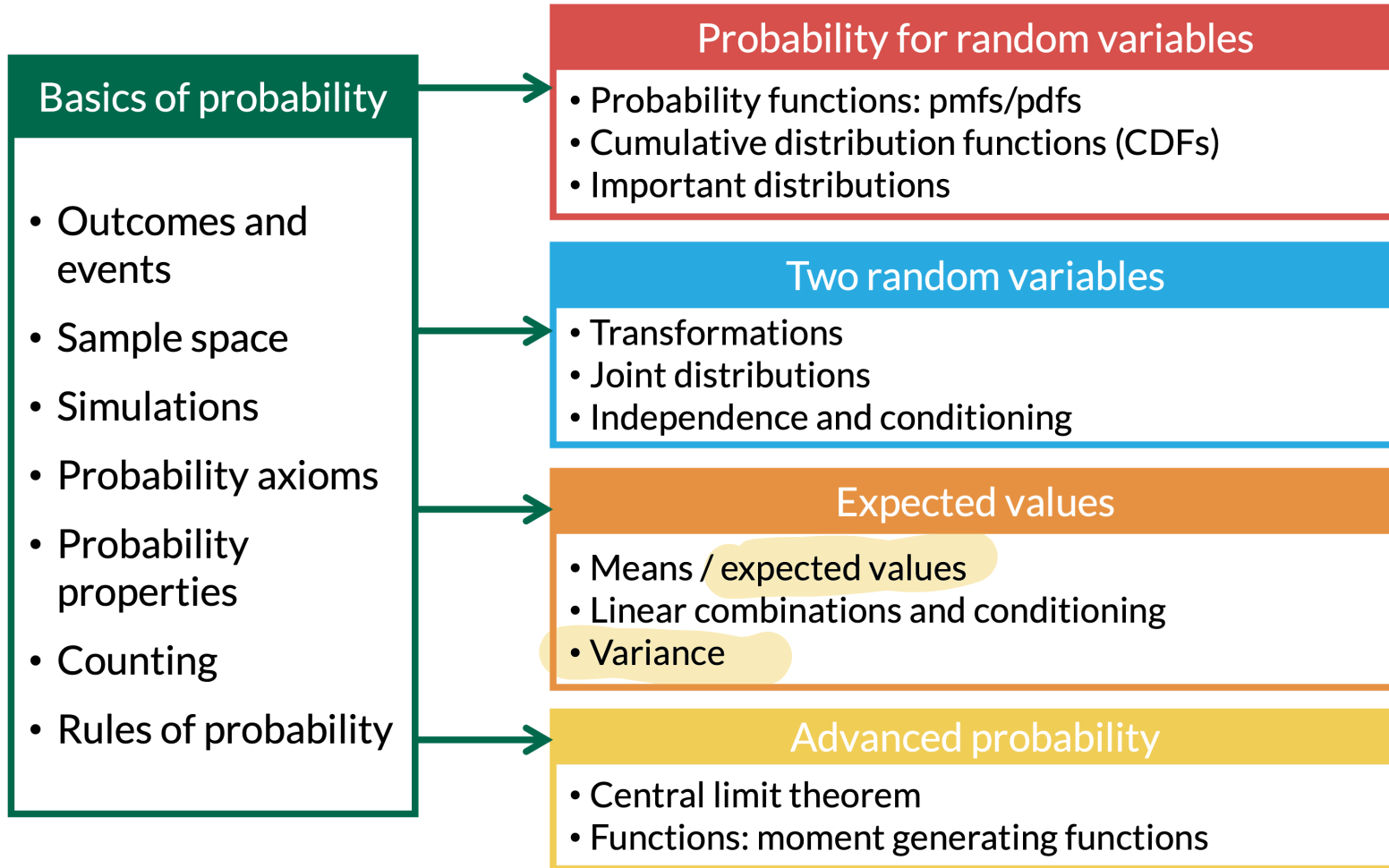
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2025-11-12

Learning Objectives

1. Define and calculate the expected value for a function of discrete and continuous RVs
2. Define and calculate variance for a single random variable
3. Define and calculate variance for multiple random variables

Where are we?



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Let's start building the variance through expected values of functions

Example 1

Let g be a function and let $g(x) = ax + b$, for real-valued constants a and b . What is $\mathbb{E}[g(X)]$?

what about $g(x) = x^2$

$$\mathbb{E}(g(X)) = \mathbb{E}(x^2)$$

$$\neq [\mathbb{E}(X)]^2$$

X is RV a & b constants

$g(X)$ is fn of RV X

$$\mathbb{E}(g(X)) = \mathbb{E}[aX + b]$$

$$= \mathbb{E}(aX) + \mathbb{E}(b)$$

$$= \mathbb{E}(a)\mathbb{E}(X) + b \quad \swarrow \text{b/c constant}$$

$$= a\mathbb{E}(X) + b$$

works b/c LINEAR fn

What is the expected value of a function? $E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$

Expected value of function of **discrete** RV

For any function g and **discrete** RV X , the expected value of $g(X)$ is

$$\mathbb{E}[g(X)] = \sum_{\{all\ x\}} g(x) p_X(x)$$

Expected value of function of **continuous** RV

For any function g and **continuous** RV X , the expected value of $g(X)$ is

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

- For example, if we have $g(x) = x^2$, then

$$\mathbb{E}[X^2] = \sum_{\{all\ x\}} \underbrace{x^2}_{\text{red circle}} \underbrace{p_X(x)}_{\text{red underline}} \neq \left(\sum_{\{all\ x\}} \underbrace{x p_X(x)}_{\text{red underline}} \right)^2 = (\mathbb{E}[X])^2$$

Let's revisit the card example (1/2)

Example 2

Suppose you draw 2 cards from a standard deck of cards with replacement. Let X be the number of hearts you draw.

1. Find $\mathbb{E}[X^2]$.

Recall Binomial RV with $n = 2$:

$$p_X(x) = \binom{2}{x} p^x (1-p)^{2-x} \text{ for } x = 0, 1, 2$$

$$p(\heartsuit) = p = 0.25$$

$$\mathbb{E}(g(X)) = \mathbb{E}(X^2)$$

$$= \sum_{\{all\ x\}} x^2 p_X(x)$$

$$= \sum_{x=0}^2 x^2 \left[\binom{2}{x} 0.25^x (1-0.25)^{2-x} \right]$$

$$= 0^2 \binom{2}{0} 0.25^0 0.75^2 + 1^2 \binom{2}{1} 0.25^1 0.75^1 + 2^2 \binom{2}{2} 0.25^2 0.75^0$$

$$\mathbb{E}(X^2) = \frac{5}{8}$$

$$\rightarrow \mathbb{E}(X^2) \neq \mathbb{E}(X)^2 = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{4}$$

Let's revisit the card example (2/2)

Example 2

Suppose you draw 2 cards from a standard deck of cards with replacement. Let X be the number of hearts you draw.

2. Find $\mathbb{E}\left[\left(X - \frac{1}{2}\right)^2\right]$.

Recall Binomial RV with $n = 2$:

$$p_X(x) = \binom{2}{x} p^x (1-p)^{2-x} \text{ for } x = 0, 1, 2$$

$$g(X) = \left(X - \frac{1}{2}\right)^2$$

$$\mathbb{E}(g(X)) = \mathbb{E}\left[\left(X - \frac{1}{2}\right)^2\right] = \sum_{\{all\ x\}} \left(x - \frac{1}{2}\right)^2 p_X(x)$$

$$= \sum_{x=0}^2 \left(x - \frac{1}{2}\right)^2 \binom{2}{x} 0.25^x 0.75^{2-x}$$

$$= \left(0 - \frac{1}{2}\right)^2 \binom{2}{0} 0.25^0 0.75^2 + \left(1 - \frac{1}{2}\right)^2 \binom{2}{1} 0.25^1 0.75^1 + \left(2 - \frac{1}{2}\right)^2 \binom{2}{2} 0.25^2 0.75^0$$

$$\mathbb{E}\left[\left(X - \frac{1}{2}\right)^2\right] = 3/8$$

ANOTHER WAY

$$\begin{aligned} \mathbb{E}\left[\left(X - \frac{1}{2}\right)^2\right] &= \mathbb{E}\left[X^2 - X + \frac{1}{4}\right] = \mathbb{E}(X^2) - \mathbb{E}(X) + \mathbb{E}\left(\frac{1}{4}\right) \\ &= \frac{5}{8} - \frac{1}{2} + \frac{1}{4} = \frac{3}{8} \end{aligned}$$

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Variance of a RV

↗ in units of X^2

Definition: Variance of RV

The variance of a RV X , with (finite) expected value $\mu_X = \mathbb{E}[X]$ is

$$\sigma_X^2 = \text{Var}(X) = \mathbb{E}[(X - \mu_X)^2] = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

$$g(X) = (X - \mu_X)^2 \\ = (X - \mathbb{E}(X))^2$$

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{n}$$

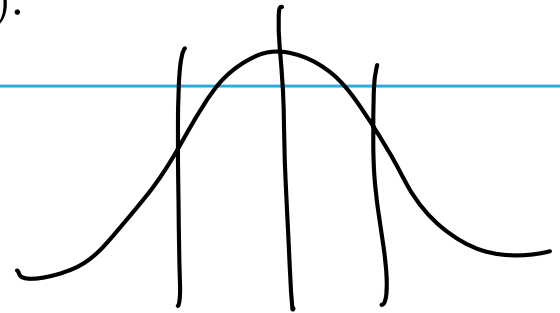
avg deviation squared

Definition: Standard deviation of RV

The standard deviation of a RV X is

$$\sigma_X = SD(X) = \sqrt{\sigma_X^2} = \sqrt{\text{Var}(X)}.$$

↘ in the units of X



Variance of discrete and continuous RVs

$$g(x) = (x - \mu_x)^2$$

How do we calculate the variance of a discrete RV?

For discrete RVs:

$$\begin{aligned} \text{Var}(X) &= \\ &= \sum_{\{all\ x\}} (x - \mu_x)^2 p_X(x) \\ &= \mathbb{E}[(X - \mu_X)^2] \\ &= \mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \end{aligned}$$

How do we calculate the variance of a continuous RV?

For continuous RVs:

$$\begin{aligned} \text{Var}(X) &= \int_{-\infty}^{\infty} (x - \mu_X)^2 f_X(x) dx \\ &= \mathbb{E}[(X - \mu_X)^2] \\ &= \mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \end{aligned}$$

Let's calculate the variance and prove it! (1/2)

$$\mu_X = E(X)$$

Lemma 6: "Computation formula" for Variance

The variance of a RV X , can be computed as

$$\begin{aligned}\sigma_X^2 &= \text{Var}(X) \\ &= E[X^2] - \mu_X^2 \\ &= E[X^2] - (E[X])^2\end{aligned}$$

$$\text{var}(X) = E((X - \mu_X)^2)$$

$$= E(X^2 - 2\mu_X X + \mu_X^2)$$

$$= \sum_{\{all\ x\}} (x^2 - 2\mu_X x + \mu_X^2) p_X(x)$$

$$\left\{ \int_{-\infty}^{\infty} (x^2 - 2\mu_X x + \mu_X^2) f_X(x) dx \right\}$$

$$= \sum_{\{all\ x\}} x^2 p_X(x) - 2 \sum_{\{all\ x\}} \mu_X x p_X(x) + \sum_{\{all\ x\}} \mu_X^2 p_X(x)$$

$$\sum_{\{all\ x\}} x p_X(x)$$

$$= \underbrace{\sum_{\{all\ x\}} x^2 p_X(x)}_{E(X^2)} - 2\mu_X \underbrace{\sum_{\{all\ x\}} x p_X(x)}_{E(X)} + \mu_X^2 \underbrace{\sum_{\{all\ x\}} p_X(x)}_{=1}$$

Let's calculate the variance and prove it! (2/2)

$$= E(X^2) - 2\mu_x E(X) + \mu_x^2$$

$$\mu_x = E(X)$$

$$= E(X^2) - 2\mu_x^2 + \mu_x^2$$

$$= E(X^2) - \mu_x^2$$

$$= E(X^2) - [E(X)]^2$$

Variance is the
(exp val of x^2) -

(the square of the
exp val)

Variance of an Uniform distribution

Example 2

Let $f_X(x) = \frac{1}{b-a}$, for
 $a \leq x \leq b$. Find $\text{Var}[X]$.

$$\begin{aligned}
 E(X^2) &= \int_a^b x^2 \frac{1}{b-a} dx \\
 &= \frac{1}{3(b-a)} x^3 \Big|_{x=a}^{x=b} \\
 &= \frac{1}{3(b-a)} (b^3 - a^3) \\
 &= \frac{(b-a)(b^2 + ab + a^2)}{3(b-a)} \\
 &= \frac{b^2 + ab + a^2}{3}
 \end{aligned}$$

last $E(X) = \frac{a+b}{2}$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$= \frac{b^2 + ab + a^2}{3} - \left(\frac{a+b}{2} \right)^2$$

$$= \frac{4(b^2 + ab + a^2)}{4(3)} - \frac{(a^2 + 2ab + b^2)3}{(4)3}$$

$$= \frac{4b^2 + 4ab + 4a^2 - (3a^2 + 6ab + 3b^2)}{12}$$

$$= \frac{b^2 - 2ab + a^2}{12} = \frac{(b-a)^2}{12}$$

$$\text{Var}(X) = \frac{(b-a)^2}{12}$$

could do:

$$\text{Var}(X) = \int_a^b (x - \mu_x)^2 \frac{1}{b-a} dx$$

Variance of exponential distribution

In the homework:

Example 3

Let $f_X(x) = \lambda e^{-\lambda x}$, for $x > 0$
and $\lambda > 0$. Find $Var[X]$.

$$\begin{aligned} Var(X) &= E(X^2) - [E(X)]^2 \\ &= \left(\int_0^{\infty} x^2 \lambda e^{-\lambda x} dx \right) - \left(\frac{1}{\lambda} \right)^2 \end{aligned}$$

∴ int by parts $2x$

$$= \frac{1}{\lambda^2}$$

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Variance of a function with a single RV

Lemma 7

For a RV X and constants a and b ,

$$g(X) = aX + b$$

$$\underline{\text{Var}(aX + b)} = a^2 \text{Var}(X).$$

Proof will be exercise in homework. It's fun! In a mathy kinda way.

Important results for *independent* RVs

Theorem 8

For independent RV's X and Y , and functions g and h ,

$$\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)].$$



Corollary 1

For independent RV's X and Y ,

$$\underline{\mathbb{E}[XY]} = \underline{\mathbb{E}[X]}\underline{\mathbb{E}[Y]}.$$

if $A \perp B$,

$$P(A \cap B) = P(A)P(B)$$

$$\mathbb{E}(g(X)h(Y)) = \sum_{\{all\ x\}} \sum_{\{all\ y\}} g(x)h(y) \underbrace{P_{X,Y}(x,y)}_{P_X(x)P_Y(y)} = \mathbb{E}(g(X))\mathbb{E}(h(Y))$$

Variance of sum of independent discrete RVs

Theorem 9: Variance of sum of independent discrete RV's

For independent ~~discrete~~ RV's X_i and constants $a_i, i = 1, 2, \dots, n$,

$$\text{Var}\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i^2 \text{Var}(X_i).$$

Simpler version:

$$\text{Var}(a_1 X + a_2 Y) = \text{Var}(a_1 X) + \text{Var}(a_2 Y) = a_1^2 \text{Var}(X) + a_2^2 \text{Var}(Y)$$

linearity of variance
for IND RVs

Corollaries

Corollary 2

For independent ~~discrete~~ ^{discrete} RV's $X_i, i = 1, 2, \dots, n$,

var of sum = sum of var

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i).$$

Corollary 3

For independent identically distributed (i.i.d.) ~~discrete~~ ^{discrete} RV's $X_i, i = 1, 2, \dots, n$,

iid

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = n \text{Var}(X_1).$$

$X_i \stackrel{\text{iid}}{\sim} N(\mu, \sigma)$

Ghost problem: *with replacement*

Example 3.2

The ghost is trick-or-treating at a different house ~~now~~. In this case it is known that the bag of candy has 10 chocolates, 20 lollipops, and 30 laffy taffies. The ghost grabs a handful of five pieces of candy. What is the variance for the number of chocolates the ghost takes? Let's solve this for the cases with replacement.

Recall probability with replacement:

binomial

$$p_X(x) = \binom{n}{k} p^k (1-p)^{n-k}$$

$X = \# \text{ chocolates (w/ rep)}$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$X = \sum_{i=1}^5 Y_i$$

$$p = \frac{10}{60} = \frac{1}{6}$$

$$Y_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p = \frac{1}{6})$$

$$P_Y(y) = p^y (1-p)^{1-y}$$

$$Y_i = \begin{cases} 1 & \text{choc} \\ 0 & \text{not choc} \end{cases}$$

$$\text{Var}(X) = \text{Var}\left(\sum_{i=1}^5 Y_i\right) = \sum_{i=1}^5 \text{Var}(Y_i) = 5 \text{Var}(Y_1)$$

$$* \text{Var}(Y_1) = E(Y_1^2) - [E(Y)]^2$$

$$\hookrightarrow E(Y) = p$$

$$E(Y^2) = \sum_{y=0}^1 y^2 P_Y(y) = 0^2 p^0 (1-p)^1 + 1^2 p (1-p)^0 = p$$

$$\text{Var}(Y_1) = p - p^2 = p(1-p)$$

$$= p$$

~~Find the mean and sd from word problem (1/2)~~

$$\begin{aligned}\text{Var}(X) &= 5 \text{Var}(Y) \\ &= 5 p \cdot (1-p) \\ &= 5 \left(\frac{1}{6}\right) \left(1 - \frac{1}{6}\right) \\ &= 0.6944\end{aligned}$$

$$\begin{aligned}\text{Var}(X) &= np(1-p) \\ &\text{Var of binomial} \\ &= n \cdot \text{Var of} \\ &\quad \text{bern}\end{aligned}$$

Back to our hotel example from Lesson 13

$$E(T) = \$6,600$$

Example 4

A tour group is planning a visit to the city of Minneapolis and needs to book 30 hotel rooms.

The average price of a room is \$200 with standard deviation \$10. In addition, there is a 10% tourism tax for each room.

What is the standard deviation of the cost for the 30 hotel rooms? Assume rooms are independent

Let T = total cost of 30 rooms

C_i = cost of room i

$$T = \sum_{i=1}^{30} 1.1 C_i$$

$$\text{Var}(T) = \text{Var}\left(\sum_{i=1}^{30} 1.1 C_i\right)$$

$$= \text{Var}\left(1.1 \sum_{i=1}^{30} C_i\right)$$

$$= 1.1^2 \text{Var}\left(\sum_{i=1}^{30} C_i\right)$$

$$= 1.1^2 \sum_{i=1}^{30} \text{Var}(C_i) = 1.1^2 \cdot 30 \text{Var}(C_1)$$

$$= 1.1^2 \cdot 30 \cdot 100 = \$^2 3,600$$

$$\text{SD}(T) = \sqrt{\text{Var}(T)} = \sqrt{\$^2 3,600} = \$60$$

WRONG

$$T = 30 \cdot 1.1 \cdot C_i$$

$$\text{Var}(30 \cdot 1.1 C_i) = (30 \cdot 1.1)^2 \text{Var}(C_i)$$

even though not IID, $\sum \text{var} = 30 \cdot \text{var}$ b/c all same var

Find the mean and sd from word problem (1/2)

Example 4

A machine manufactures cubes with a side length that varies uniformly from 1 to 2 inches. Assume the sides of the base and height are equal. The cost to make a cube is 10 ¢ per cubic inch, and 5 ¢ cents for the general cost per cube. Find the mean and standard deviation of the cost to make 10 cubes.

$$\begin{aligned} E(X_i^3) &= \int_1^2 x_i^3 \cdot 1 \, dx \\ &= \frac{1}{4} x_i^4 \Big|_{x_i=1}^{x_i=2} \\ &= \frac{1}{4} (2^4 - 1^4) = \frac{15}{4} \end{aligned}$$

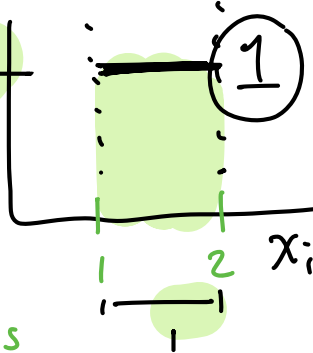
Model cost for 10 cubes:

Let C = cost of 10 cubes (in cents)

C_i = cost of cube i ($i=1, 2, \dots, 10$)

X_i = length of each side of cube i

$$C = \sum_{i=1}^{10} C_i \quad C_i = 5 + 10(X_i^3) \quad f(x)$$



MEAN $E(C) = E\left[\sum_{i=1}^{10} (5 + 10X_i^3)\right]$

$= \sum_{i=1}^{10} E(5 + 10X_i^3)$ (linear combos)

$= \sum_{i=1}^{10} (5 + 10E(X_i^3))$ (constants 5 & 10)

$= \sum_{i=1}^{10} (5 + 10(\frac{15}{4})) = \sum_{i=1}^{10} 42.5 = 425 \text{ ¢}$

$E(C) = 425 \text{ ¢}$

Find the mean and sd from word problem (1/2)